

Assignment_1: The LP Model

1. **Back Savers:** This is a company that produces two types of backpacks for students - the Collegiate and the Mini. The management wants to know how much we should produce each type of backpack per week.

a. Clearly define the decision variables:

- $C = \text{number of Collegiate produced per week}$
- $M = \text{number of Mini produced per week}$

b. What is the objective function?

- $\text{Max Profit (X)} = 32C + 24M$
- Here, X indicates the maximum amount of profit we will get.

c. What are the constraints?

$C = \text{no. of collegiate produced per week}$

$M = \text{no. of Mini produced per week}$

Here, we have to keep the production number that will give us a max profit X

$$\text{Max profit (X)} = 32C + 24M$$

Now, we will create a different constraint:

Nylon Material Constraint:

Each Collegiate backpack requires 3 sqft of nylon and each Mini backpack requires 2 sqft. A total of 5000 sqft of nylon is available and we should use equal to or less than 5000 sqft but not more than it.

$$(3C + 2M \leq 5000)$$

Collegiate Sales Constraint:

At Most 1000 collegiate backpacks can be sold per week.

$$(C \leq 1000)$$

Mini Sales Constraint:

At most 1200 mini backpacks can be sold per week.

$$(M \leq 1200)$$

Labor Constraint:

There are 35 laborers, and each laborer works 40 hours per week.

$$(35 * 40 = 1400) \text{ labor hours per week}$$

Converting labor hours to minutes:

$$(1400 * 60 = 84000 \text{ minutes})$$

Now, each collegiate backpack requires 45 minutes of labor and each Mini backpack requires 40 minutes of labor. So, we formulate that like this

$$(45C + 40M \leq 84000)$$

Lastly, we create Non-Negativity Constraint as well because C and M represent quantities of backpack produced, they cannot be negative for any reason.

$$(C \geq 0, M \geq 0)$$

d. Full Mathematical Formulation:

$$(\text{Maximize}) X = 32C + 24M$$

Subject to:

$$3C + 2M \leq 5000$$

$$C \leq 1000 \text{ and } M \leq 1200$$

$$45C + 40M \leq 84000$$

$$C, M \geq 0$$

2. **The Weigelt Corporation:** Here management wants to know how many units of each size should be produced at each plant to maximize profit.

a. Define the decision variables:

We have 3 size and 3 plants,

So,

$L1 = \text{no. of large units production at plant 1 per day}$

$L2 = \text{no. of large units production at plant 2 per day}$

$L3 = \text{no. of large units production at plant 3 per day}$

$M1 = \text{no. of medium units production at plant 1 per day}$

$M2 = \text{no. of medium units production at plant 2 per day}$

$M3 = \text{no. of medium units production at plant 3 per day}$

$S1 = \text{no. of small units production at plant 1 per day}$

$S2 = \text{no. of small units production at plant 2 per day}$

$S3 = \text{no. of small units production at plant 3 per day.}$

b. Formulate a linear programming model for this problem.

Objective Function:

The profit per unit is:

- Large = \$420
- Medium = \$360
- Small = \$300

Since each size can be produced at all three plants, total profit will be:

$$(\text{Max Profit } (X) = 420(L1 + L2 + L3) + 360(M1 + M2 + M3) + 300(S1 + S2 + S3))$$

Production Capacity Constraints:

Plant 1 can produce at most 750 units per day,

$$(L1 + M1 + S1 \leq 750)$$

Plant 2 can produce at most 900 units per day,

$$(L2 + M2 + S2 \leq 900)$$

Plant 3 can produce at most 450 units per day,

$$(L3 + M3 + S3 \leq 450)$$

Storage Space Constraints:

Plant 1 has 13000 sqft of available storage space:

$$(20L1 + 15M1 + 12S1 \leq 13000 \text{ sqft})$$

Plant 2 has 12000 sqft of available storage space:

$$(20L2 + 15M2 + 12S2 \leq 12000 \text{ sqft})$$

Plant 3 has 5000 sqft of available storage space:

$$(20L3 + 15M3 + 12S3 \leq 5000 \text{ sqft})$$

Sales Demand Constraints:

The total no. of large units produced across all three plants cannot exceed 900 units per day.

$$(L1 + L2 + L3 \leq 900 \text{ units per day})$$

The total no. of Medium units produced across all three plants cannot exceed 1200 units per day

$$(M1 + M2 + M3 \leq 1200 \text{ units per day})$$

The total no. of Small units produced across all three plants cannot exceed 750 units per day.

$$(S1 + S2 + S3 \leq 750 \text{ units per day})$$

Now,

Management wants all three plants to use the same percentage of their available excess production capacity.

Therefore,

$$(L1 + M1 + S1 / 750) = (L2 + M2 + S2 / 900) = (L3 + M3 + S3 / 450)$$

Now,

$$A = L1 + M1 + S1$$

$$B = L2 + M2 + S2$$

$$C = L3 + M3 + S3$$

So,

$$A/750 = B/900$$

Cross-multiplying gives:

$$900A = 750B$$

Substituting the variable back in:

$$900(L1 + M1 + S1) = 750(L2 + M2 + S2)$$

Now lets compare Plant 2 and 3:

$$B/900 = C/450$$

Cross-multiplying gives:

$$450B = 900C$$

So,

$$450(L2 + M2 + S2) = 900(L3 + M3 + S3)$$

Lastly, we create a non-negative constraint, because all 9 decision variables represent the number of units produced, none of them can be negative.

$$(L1, L2, L3, M1, M2, M3, S1, S2, S3 \geq 0)$$